

Business Forecasting Project 2 Final Report

Sales Forecasting Framework for Retail Operations

Business Forecasting (WIHN0032)
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1. Introduction

This project aims to generate 6 week sales forecasts for 676 stores of a European drugstore chain. We first performed data preprocessing and exploratory data analysis (EDA) to understand sales patterns and identify important influencing factors. Statistical benchmark models were then established to provide baseline performance for comparison. Finally, a global Random Forest forecasting framework was developed by integrating historical sales, engineered time series features, store metadata, and operational variables. Since future customer counts are unavailable but strongly related to sales, a customer forecasting model was first constructed, and its predictions were incorporated into the final sales forecasting model.

The forecasting performance of the proposed framework is evaluated and compared with benchmark models to identify an accurate and practical solution for large scale retail forecasting.

2. Data Understanding

Three datasets were provided for this project: sales, future, and metadata. Together, these datasets contain historical sales records and future operational information.

The historical sales dataset consists of 624,624 daily observations collected from 676 stores, covering the period from 07.01.2013 to 19.07.2015. Each record contains daily sales, customer counts, store operating status, promotional activities, and holiday information. The future values dataset provides known future operational variables for the forecasting period from 20.07.2015 to 17.09.2015, while the target variable (sales) is intentionally unavailable because it represents the forecasting objective. The metadata dataset contains static information, including store type, assortment level, and competition distance.

Although future data contains 60 future dates, the project requires a 6 week horizon. Therefore, only the first 42 dates were used for the final forecast and business analysis. These three datasets complement each other by combining historical sales record, future operational conditions and some store characteristics, providing the necessary information for building a 6 week sales forecasting framework.

3. Exploratory Data Analysis

To understand the dataset's characteristics and identify key drivers of sales variance, an Exploratory Data Analysis (EDA) was initially conducted. This analysis serves as the foundation for subsequent model construction and feature engineering.

First, an examination of the daily sales distribution across all stores revealed pronounced differences, indicating that each store exhibits distinct sales patterns and demand profiles. Therefore, when developing predictive models, it is essential to include different characteristics of all stores alongside historical sales data, rather than relying only on aggregate or average historical behaviors.

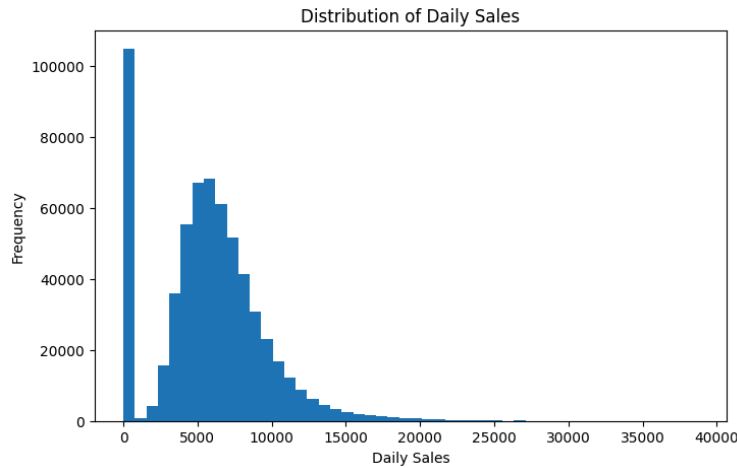


Fig.1 Distribution of Daily Sales

Second, the impact of promotional activities on sales was evaluated. The results indicated that the average sales during promotional periods were significantly higher than those during no promotional periods. This demonstrates that promotions are a critical external factor driving demand, thereby justifying the retention of the promo variable as an important predictor in subsequent modeling stages.

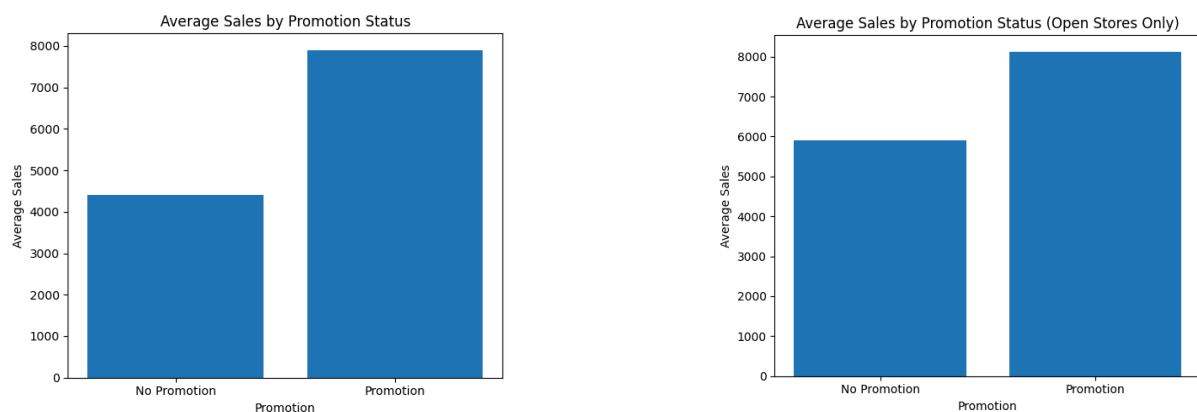


Fig. 2 Average Sales by Promotion Status

Furthermore, the influence of holidays on sales was analyzed. During national and festival holidays, sales in some stores dropped close to zero, primarily driven by scheduled store closures. Consequently, state holiday and open serve as effective indicators for capturing sudden shifts in demand, offering substantial predictive value for sales forecasting.

Given that customer traffic directly reflects store demand, this study also investigated the relationship between the number of customers and total sales. The analysis revealed a strong positive correlation between the two metrics (correlation = 0.8822), establishing customer traffic as a primary driver of sales performance. Since future customer counts are

unavailable during the actual forecasting horizon, introducing a predicted customer variable into the sales model risks cascading forecasting errors from the customer model into the final sales predictions. To mitigate this risk, the baseline benchmark phase excluded the customer variable. Nonetheless, during the Random Forest modeling phase, we made two separate versions: one incorporating and one omitting the customer feature, to systematically evaluate its impact and examine the resulting differences.

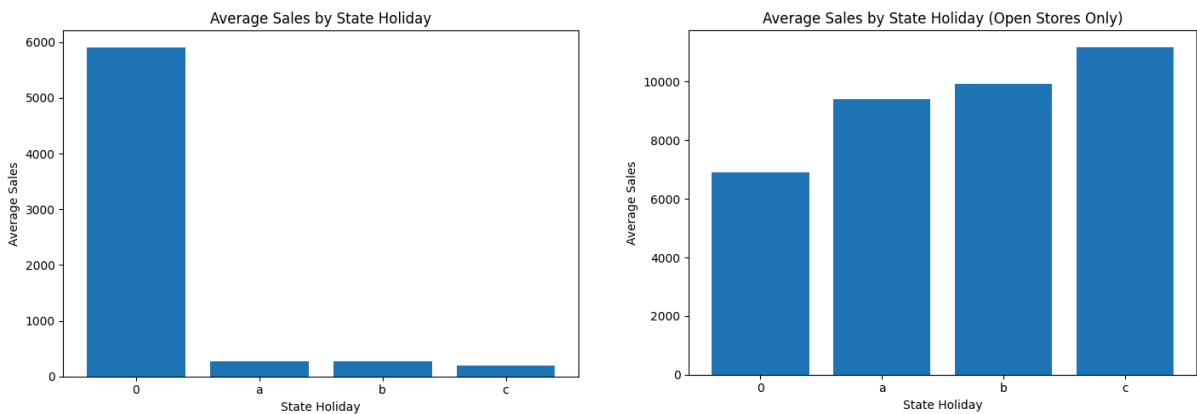


Fig.3 Average Sales by State Holiday

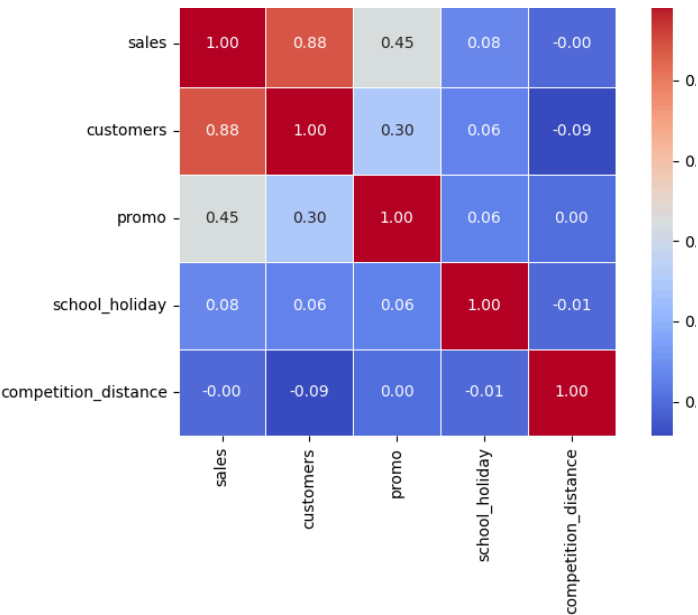


Fig.4 Correlation Matrix of Important Variable and Sales

We also observed the temporal dynamics of daily sales, which uncovered a distinct weekly seasonality characterized by stable demand fluctuations across different days of the week. To capture these time dependent patterns, various time series features were integrated into the subsequent models, including lag features, rolling statistics, and calendar features.

In summary, the EDA demonstrates that sales are not merely a function of past sales volumes; rather, they are jointly shaped by promotional activities, holiday effects, store operational statuses, customer demand, temporal cycles, and inherent specific store

characteristics. These analytical insights directly inform the design of the subsequent feature engineering pipeline and forecasting models.

4. Data Preparation

4.1 Data Cleaning

Before model development, the three raw datasets were examined for missing values, duplicate observations, data type inconsistencies, and store coverage. The historical sales dataset contained 624,624 observations for 676 stores and had no missing values in any of its original variables and duplicated rows. The future dataset contained 40,560 observations, corresponding to 60 future dates for all 676 stores. It also contained no duplicates. The metadata file contained one record for each of the 676 stores, with no duplicates.

Three types of missing values were identified. First, the open variable contained 11 missing observations in the future dataset. All 11 records belonged to store_379 and occurred on weekdays or Saturdays. Inspection of the store's recent historical operating pattern showed that it was normally open on these days and closed on Sundays. Therefore, these 11 values were imputed as 1, indicating that the store would be open.

Second, competition distance contained one missing value in the metadata. The observed values ranged from 20 to 75,860 metres and were positively skewed, with a median of 2,640 metres. Because only one value was missing and the median is less sensitive to extreme distances than the mean, the missing value was replaced with the median.

Third, all 40,560 values in the future customers column were missing. These values were not treated as ordinary errors because future customer numbers were genuinely unavailable at the time of forecasting. Customer demand was handled separately by developing a customer forecasting model, whose predictions were later used as inputs to the final sales model (in one of Random Forest Model).

Date variables were converted to datetime format, while categorical variables such as store_id, state_holiday, store_type, and assortment were assigned appropriate categorical data types. Store metadata were then merged into both the historical and future datasets using store_id. After cleaning and merging, no missing values remained except for the intentionally unavailable future customer counts.

4.2 Feature Engineering

Preparing for Random Forest Model, we did some serious data cleaning and feature engineering. First, we sorted everything strictly by store ID and date to keep the timeline intact, which ensures our time series features are accurate. We also renamed the key columns, changing store id to unique id, date to ds, and sales to y, to match what the ML forecasting package expects. For the categorical variables, we used mapping dictionaries to turn them into integers, state_holiday values (0, a, b, c) became 0, 1, 2, 3; store_type (a, b, c, d) became 0, 1, 2, 3; and assortment (a, b, c) became 0, 1, 2.

When it came to setting up the time and contextual features, we used a mix of manual engineering and package tools to capture different angles of the timeline. We manually created features for quarter week of year, and also multiplied the promo and weekend

variables to create an interaction feature called `promo_weekend`. Core calendar features like month were left to the forecasting package to extract automatically during training.

To really capture the trends and historical momentum in the time series, we set up key lag features for the final sales model spanning a bit over four weeks, including 1, 7, 14, 21, and 28 day lags. At the same time, we built two rolling average features: a 4 day rolling mean based on lag 7 (`RollingMean(window_size=4)`), and a 2 day rolling mean based on lag 28 (`RollingMean(window_size=2)`). This helps the model balance short term ups and downs with smoother long term baselines.

We also tried integrating historical actual customers with future customer predictions. From there, we generated extra features like `customer_promo` (predicted customers during promos), `customer_weekend` (predicted customers on weekends), and `customer_ratio` (the ratio of daily predicted customers to the store's historical average). Feeding these highly relevant features into a single Global Random Forest model allowed it to learn overarching patterns across all stores at once.

5. Benchmark Models

As the baseline models are based on univariate statistical forecasting, it is assumed that historical sales data already captures the primary information required to predict future sales. Therefore, introducing external variables, such as promotions, holidays, is unnecessary here.

Since the benchmark models were designed as simple reference baselines, they were evaluated using a 42 day holdout test set instead of cross validation. This considerably reduced computational cost while providing a consistent baseline for comparison. Rolling validation was applied only to the Random Forest models.

5.1 Benchmark Model Selection

For this project, we selected Mean, Naive, Drift, Seasonal Naive, AutoARIMA, and ETS as our benchmark models. We want to set up a baseline ranging from simple to more advanced time series models. This helps us see what kind of forecasting performance we can get just by using each store's historical sales data. If Random Forest can't consistently beat these benchmarks, it means the extra complexity of feature engineering and machine learning might not actually be worth it.

We treat each store as an independent time series and train them on their own historical sales data. We used the last 42 days as the test period, and everything before that as the training set. All models use the exact same train/test split and are evaluated using primarily MAPE and MAE, ensuring a fair and consistent comparison across the board.

Why we use these models:

- Mean: It's simple, fast to compute, and provides a stable and computationally inexpensive long run level baseline, although it can still be influenced by extreme observations.
- Naive: The main perks are that it's straightforward, takes almost zero computing power, and carries over the most recent sales level.

- Drift: This builds a baseline with a simple trend, helping us see if adding a linear direction performs better than just using a flat average or the last data point.
- Seasonal Naive: Based on our EDA, we noticed strong weekly patterns. This model sets up a baseline using only weekly seasonality, letting us test how much just recognizing a 7 day cycle improves the forecast.
- AutoARIMA: This provides a more comprehensive, auto traditional time series benchmark that includes weekly seasonality. We use it to see if a more complex statistical model clearly outperforms simple rules of thumb.

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Mean and Median Accuracy by Model
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	MAE_Mean	MAE_Median	MAPE_Mean	MAPE_Median
model				
Mean	2126.523	2039.056	20.748	19.939
AutoARIMA	2330.696	1744.770	31.556	21.150
Seasonal Naive	2146.890	2106.440	39.753	42.666
Naive	5795.228	5588.250	97.778	100.000
Drift	6025.591	5830.975	101.232	103.453

Fig. 5 Table of Mean and Median Accuracy by Model

After reviewing the initial results, we decided to add ETS as another benchmark model.

- ETS: As the results of metrics were not very ideal, and AutoARIMA also takes a long time to run searches across all stores, we brought in ETS to balance speed and accuracy. According to EDA Step 18 “Weekly and Month Sales Pattern”, our exploratory analysis showed a clear weekly pattern in the overall data, so we set the ETS seasonal period to 7 days. Even though the aggregated data doesn't show a glaring global trend, individual stores might still have their own local trends, so we kept the additive trend component and will validate whether it fits through our benchmark comparison. The actual configuration is set to additive trend, additive seasonality, and seasonal_periods=7. All stores use the same ETS specification: additive trend, additive seasonality, and a seasonal period of seven days. But the model parameters and state components are estimated separately for each store.

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ETS Accuracy Summary
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	MAE_Mean	MAE_Median	MAPE_Mean	MAPE_Median
model				
ETS	1039.996	1003.901	17.071	16.304

Fig. 6 Table of Mean and Median Accuracy for ETS

5.2 Benchmark Model Results

Because the sales scale varies across different stores, a percentage error metric is much better suited for cross store comparisons. MAPE was calculated only for observations with non-zero actual sales. For the benchmark models, MAPE was first calculated separately for each store and subsequently averaged across stores.

5.2.1 Overall Results of the Initial Benchmark Models

Among the five initial models, the Mean model had the lowest average MAPE at 20.75%, making it the best performing benchmark overall before ETS was introduced. Even though Seasonal Naive's average MAE (2,146.89) was close to Mean's (2,126.52), its average MAPE reached 39.75%, which is significantly higher than the Mean model. This indicates that while Seasonal Naive's absolute errors might not be, but the relative error remains high when measured against each store's actual sales volume.

AutoARIMA delivered an average MAPE of 31.56%. While it beat Seasonal Naive, Naive, and Drift, it failed to outperform the simpler Mean model. This shows that even though AutoARIMA can automatically identify time dependent structures and factor in a weekly periodicity seasonality, its higher model complexity did not translate into a lower overall average forecasting error.

Naive and Drift performed significantly worse, showing that models that forecast by simply extending trends do not match our sales patterns. In contrast, ETS became the top performing benchmark model overall. Its average MAPE was 17.07%, lower than the previous best performer (Mean model at 20.75%). Its average MAE was 1,039.996, which was also noticeably lower than the Mean model's 2,126.523. The average MAPE of ETS is about 15% lower than that of the Mean model (in terms of relative reduction in Mean and Median of MAPE). Furthermore, the average MAE of ETS is about 1,086.53 sales units lower than the Mean model, representing roughly a 51.1% drop (in terms of relative reduction in Mean and Median of MAE). Both metrics provide that ETS's advantage in percentage calculation, but also reflects a genuine reduction in actual sales volume errors.

5.2.2 Results by Store Type

To verify whether the model only performs well on specific types of stores, we merged the benchmark results with the store_type from the metadata, and calculated the average MAE and MAPE for store types a, b, c, and d separately. For each type, the model with the lowest average MAPE was chosen as the best model for that category.

The results show that ETS achieved the lowest average MAPE across all four store types. Therefore, ETS's overall advantage isn't driven by just one specific store type; it maintains the best average percentage error across categories a, b, c, and d.

Specifically, type b stores had the lowest ETS average MAPE at 11.93%, followed by type a at 16.52%, type d at 17.77%, and type c at 18.61%. However, there are only 11 stores in type b, far fewer than type a (377 stores), type c (75 stores), and type d (213 stores). As a result, its average results represent a much smaller sample size. Based on the current output, we can confirm that ETS ranks first across all types, but we cannot conclude that the differences between these types are statistically significant based on these averages alone. We identified the model with the lowest MAPE for each individual store, and then calculated the win rate for each model. The results are as follows:

Model	Stores Won	Win Rate (%)
ETS	449	66.42
Mean	135	19.97
AutoARIMA	88	13.02
Seasonal Naïve	4	0.59

Fig. 7 Ratio of Best Model

A total of 676 stores were compared. ETS achieved the lowest MAPE in 449 of these stores, translating to a win rate of 66.42%, meaning ETS performed the best in roughly two-thirds of the stores. This outcome supports our overall average comparison: ETS is actually winning across the majority of individual stores.

Summary

Considering the overall average performance, the results by store type, and the individual store win rates, ETS stands out as the strongest benchmark model for this project:

- ETS has the lowest overall average MAPE among all models.
- ETS has the lowest overall average MAE at 1,039.996.
- ETS is the model with the lowest average MAPE across all four store types.
- ETS came out on top in 449 out of 676 stores, achieving an individual store win rate of 66.42%.

We decide not to do more steps of cross validation. Not just for saving time, also because this evaluation strategy is consistent with the project objective of forecasting the next 6 weeks. Model robustness was assessed from multiple perspectives, including overall average MAPE, performance across all four store types, and the best performing model for each individual store. ETS consistently achieved the lowest average MAPE, ranked first across all store types.

Consequently, ETS should serve as the primary baseline for comparison with our upcoming Random Forest model. The Mean model and AutoARIMA still offer some reference value, as they achieved the best results in 19.97% and 13.02% of the stores, respectively. However, if we need to choose a single model to represent our overall benchmark performance, all evaluation metrics point to ETS.

5.3 ETS Forecast Result

We used ETS as our Final Benchmark Forecast Result and saved the sales predictions for each store into a CSV file for future use. The Total Sales predicted by ETS are shown below.


```
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ETS Final Forecast
=====
Stores forecasted      : 676
Forecast days         : 42
Total forecast rows    : 28392
Total sales result     : 173,830,086.84
```

Fig. 8 ETS Sales Forecasting Result

The results are consistent with the EDA finding that weekly seasonal structure is useful for forecasting. Moving forward, we should place a heavier emphasis on weekly cycles when making business decisions. Based on the findings from this timeframe, ETS is the most suitable model for us when we need to complete sales evaluations quickly due to time or cost constraints.

6. Random Forest Model

6.1 Baseline Global Random Forest

Before developing the final forecasting framework, a Baseline Global Random Forest was constructed as the starting point for model development. Unlike the statistical benchmark models presented before, this baseline serves as a reference for evaluating the contribution of subsequent modelling improvements.

The baseline model assumes that future sales can be explained using historical sales together with future known operational variables. Customer information is intentionally excluded to establish a reference model for subsequent comparison.

The model was trained on historical data from all stores simultaneously using historical sales lag features, rolling statistics, calendar variables, promotional and holiday information, and static store characteristics.

The baseline model followed the same training, hyperparameter tuning, and validation procedures as the final forecasting framework. The only difference was the exclusion of customer information. This model achieved a MAPE of 10.12%, providing a reference for assessing the effectiveness of the proposed two stage forecasting framework.

6.2 Customer-Augmented Forecasting Framework

Although the Baseline Global Random Forest incorporated historical sales, calendar variables, and store characteristics, its forecasting accuracy indicated room for improvement. Exploratory analysis showed that customer traffic is strongly associated with sales, suggesting that customer information could further improve forecasting performance.

Since future customer counts are unavailable at the forecasting stage, a two stage forecasting framework was developed. This approach assumes that future customer demand can be forecast with sufficient accuracy and used as an explanatory variable for sales forecasting. First, a Global Random Forest model was trained to forecast customer counts for the next 42 days using recursive forecasting. These predicted customer counts were then incorporated into the sales forecasting model together with future known variables, including

promotional activities, store opening status, holiday information, and static store characteristics.

To capture customer counts more effectively, several customer features, including predicted customer counts (customer_prediction), customer ratio, customer promo, and customer weekend, were included in the sales model. The proposed forecasting framework assumes that the historical relationship between customer counts and sales remains sufficiently stable over the 6-week forecasting horizon. During training, historical customer observations were used, while during future forecasting they were replaced by the customer forecasts generated in the first stage, preventing data leakage while preserving the predictive value of customer demand.

By incorporating customer forecasts, the average MAPE was reduced from 10.12% to 7.82%, demonstrating the effectiveness of the proposed two stage Global Random Forest framework.

Stage	Model	MAPE (%)	Purpose
Baseline	Global Random Forest	10.12	Baseline machine learning model
Enhancement	Global Random Forest + Customer Forecast	7.82	Evaluate the contribution of customer forecasting

Table 1. Comparison of Baseline and Enhancement Random Forest Model

6.3 Hyperparameter Selection and Model Validation

To balance forecasting accuracy and model complexity, two Random Forest configurations were evaluated for both the customer and sales forecasting models. The candidate models differed in the number of trees, minimum leaf size, and the maximum number of features considered at each split.

All models were trained using the same feature engineering pipeline and evaluated using Rolling Forecast Origin Cross Validation. The forecasting horizon was fixed at 42 days, matching the required 6 week prediction period. This validation strategy preserves the chronological order of the data and prevents future information leakage. To improve computational efficiency, `refit = False` was adopted, meaning that the model was not retrained at every forecasting origin while maintaining a consistent evaluation framework.

Forecasting performance was evaluated using MAPE and MAE. MAPE served as the primary hyperparameter selection criterion because it enables comparisons across stores with different sales scales.

The table summarises the hyperparameter tuning results. For both forecasting tasks, RF200_SQRT achieved the lowest MAPE and was therefore selected as the final configuration. Increasing the number of trees and the minimum leaf size did not improve forecasting accuracy. Consequently, both forecasting models adopted the same hyperparameter settings of 200 trees, `max_features = sqrt`, and `min_samples_leaf = 1`.

Forecasting Task	Model	Trees	Max Features	Min Leaf	MAPE (%)	Selected
Customer	RF200_SQRT	200	sqrt	1	8.63	✓
Customer	RF300_SQRT_Leaf5	300	sqrt	5	8.77	
Sales	RF200_SQRT	200	sqrt	1	7.82	✓
Sales	RF300_SQRT_Leaf5	300	sqrt	5	8.04	

Table 2. Model Selection

7. Results

7.1 Benchmark Comparison and Model Improvement

To evaluate the proposed forecasting framework, the final Global Random Forest model was compared with ETS, the best benchmark model identified .(MAPE = 17.07%).

Replacing ETS with a Global Random Forest reduced the MAPE to 10.12%. Incorporating predicted customer counts and customer features further reduced the MAPE to 7.82%.

Overall, the proposed framework reduced the average MAPE by 54.2% and the average MAE by 45.4% compared with ETS.

The improvement in forecasting accuracy can be attributed to two factors. First, the Global Random Forest captures complex nonlinear sales patterns more effectively than the statistical ETS model. Second, incorporating customer forecasts provides additional information about future demand, further improving forecasting accuracy. This is supported by the feature importance analysis, where customer variables rank among the most influential predictors.

Model	MAE	MAPE	Improvement
ETS	1039.996	17.07	–
Final Global RF	567.370	7.82	↓54.2% MAPE

Table 3. Comparison of ETS and Final Global Model

7.2 Forecast Accuracy Analysis

7.2.1 Accuracy by Store Type

Forecasting accuracy is generally consistent across the four store types, with MAPE values ranging from 7.55% to 7.90% for Types A, C, and D. Type D achieved the lowest forecasting error (7.55%), while Type C and Type A showed similar performance. In contrast, Type B recorded a higher MAPE (11.24%). However, this result should be interpreted with caution because Type B contains only 1,386 validation samples, substantially fewer than the other store types. Overall, the model demonstrates stable forecasting performance across different store types without showing a systematic bias toward any particular category.

7.2.2 Accuracy by Sales Volume

Forecasting performance is consistently good across all sales groups, with MAPE values below 9%. The Medium sales group achieved the highest accuracy (7.27%), followed by the High sales group (7.68%), while the Low sales group recorded a slightly higher MAPE (8.52%). This suggests that stores with lower sales are relatively more difficult to forecast, as small absolute deviations correspond to larger percentage errors. Nevertheless, the differences between the three groups are modest, indicating that the proposed framework generalises well across stores with different sales volumes.

7.2.3 Accuracy by Forecast Horizon

Forecasting accuracy remains relatively stable throughout the 6 week forecasting horizon, with all weekly MAPE values below 10%. The highest accuracy is achieved during Week 2 (6.38%), followed closely by Week 1 (6.62%). As expected, forecasting errors generally increase for longer horizons, with Week 3 showing the highest MAPE (9.05%), while Weeks 5 and 6 remain below 9%.

This pattern is consistent with the nature of time series forecasting, where uncertainty accumulates as predictions extend further into the future. Despite this increase, the model maintains relatively stable performance throughout the entire 6 week period, with only moderate variation between weeks. These results indicate that the proposed forecasting framework is sufficiently robust for both short term operational planning and medium term business decision making.

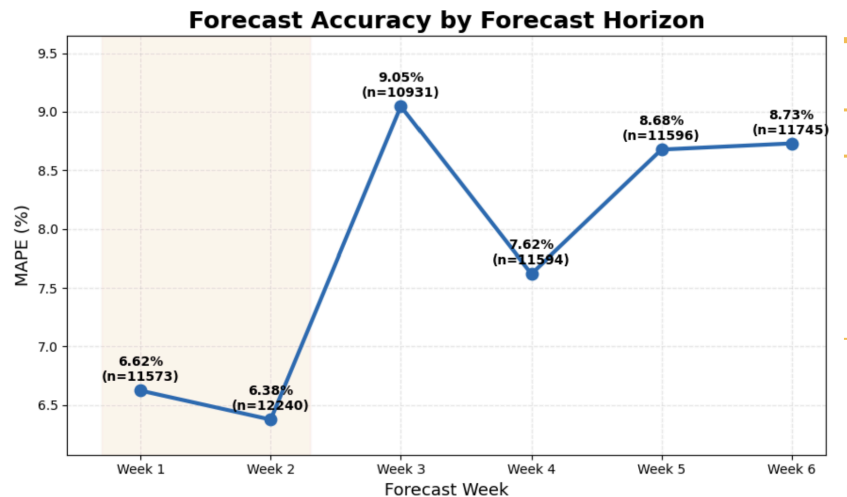


Fig 9. Forecast Accuracy

7.2.4 Accuracy by Promotion

Forecasting accuracy is similar under both promotional and non promotional conditions. The model achieved a MAPE of 7.77% for non promotion periods and 7.89% during promotions, with a difference of only 0.12 percentage points. This indicates that promotional activities do not substantially reduce forecasting accuracy, suggesting that the model effectively captures the impact of promotions through the inclusion of promotion features.

7.3 Feature Importance

To improve the interpretability of the final forecasting model, Permutation Feature Importance was applied to evaluate the contribution of each input feature. Feature importance was measured by randomly permuting a feature and observing the resulting increase in forecasting error.

The results show that the model combines customer information, historical sales patterns, and operational variables, rather than relying on a single source of information.

Among all variables, customer ratio and customer count are the two most important features, highlighting the dominant role of customer demand in predicting future sales. In the implementation, customer count is displayed in the figure for readability but corresponds to the variable customer prediction, which contains historical customer observations during training and predicted customer values during forecasting. Its high importance therefore reflects the predictive value of customer demand rather than the use of future information.

Historical sales features, including lag1, lag14, lag28, and the rolling mean, are also ranked among the top ten, demonstrating that the model captures both recent sales dynamics and recurring temporal patterns. In addition, open, promo, and customer promo contribute substantially by modelling store operations and the interaction between promotions and customer demand.

Overall, the feature ranking aligns well with retail business logic and provides strong support for the proposed two stage forecasting framework. The dominance of related customer features also explains why incorporating customer forecasts reduced the MAPE from 10.12% in the baseline model to 7.82% in the final model.

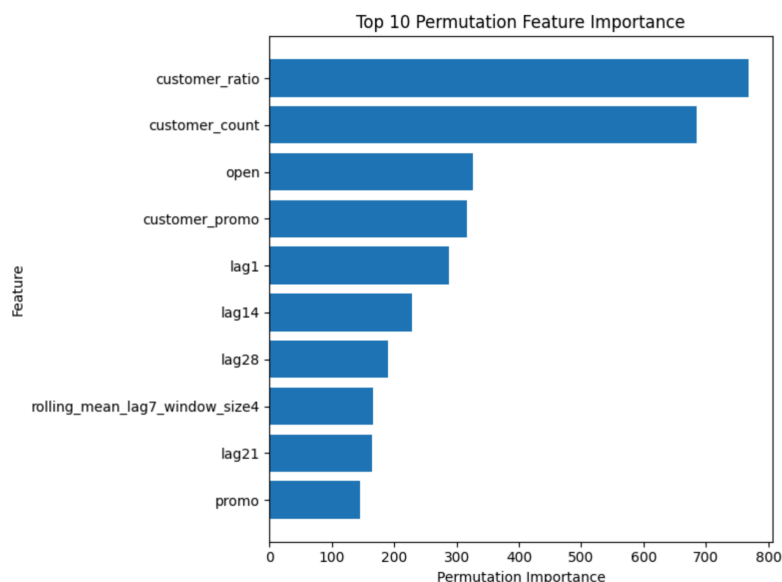


Fig 10. Top 10 Permutation Importance

7.4 Business Decision Support

7.4.1 Strategic Decision Support (CFO)

For the CFO, the total sales expected over the 6 week planning horizon represent the most important business indicator, as they directly support cash flow management, budgeting, and procurement planning. The proposed forecasting framework estimates total company sales of 164.9 million over the next six weeks, providing a clear estimate of future revenue and a solid basis for short term financial planning.

The cumulative sales forecast illustrates how revenue is expected to accumulate throughout the 6 week forecasting horizon. The steady increase in cumulative sales indicates a stable revenue stream without significant interruptions, allowing the finance department to anticipate future cash inflows and schedule expenditures more effectively.

The company level forecast reveals a clear weekly sales cycle throughout the forecasting period, with recurring peaks and troughs that are consistent with historical retail demand patterns. At the aggregated company level, random fluctuations observed in individual stores are largely smoothed out, making the forecast more reliable for financial planning, budgeting, and revenue forecasting in overall condition.

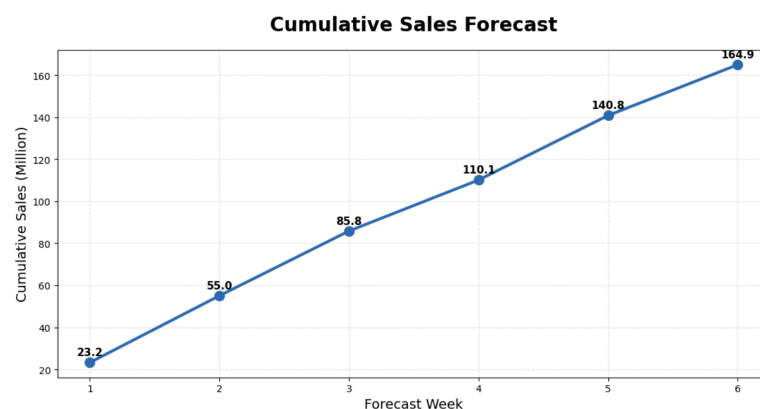


Fig 11. Cumulative Sales Forecast

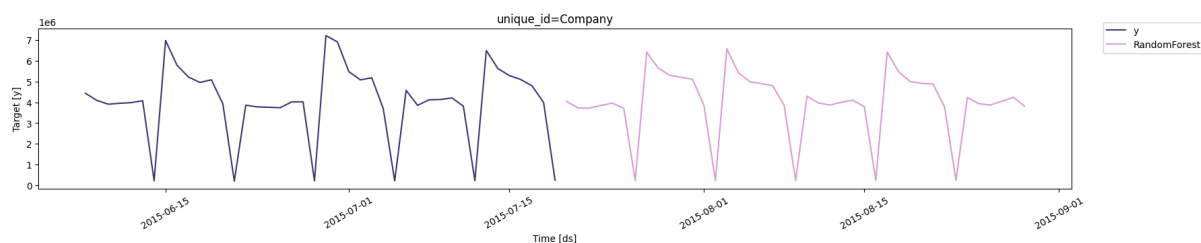


Fig. 12 Weekly Sales Forecast

Together, the three business outputs provide complementary information for financial decision making. The 6 week total sales KPI summarises the expected revenue, the cumulative sales forecast illustrates the expected cash flow accumulation, and the company level forecast explains the underlying revenue dynamics. These forecasts provide valuable support for budgeting, cash flow management, and procurement planning.

7.4.2 Operational Decision Support (COO)

COO is primarily concerned with short term operations. Therefore, the analysis focuses on the first two forecast weeks, providing information for inventory replenishment, logistics planning, and workforce scheduling.

To identify operational priorities, the top twenty stores with the highest forecast sales were analysed. Although these stores represent only a small proportion of all locations, they contribute approximately 6.7% of the company's forecast sales. This highlights the importance of prioritising high performing stores when allocating inventory, delivery capacity, and staffing resources.

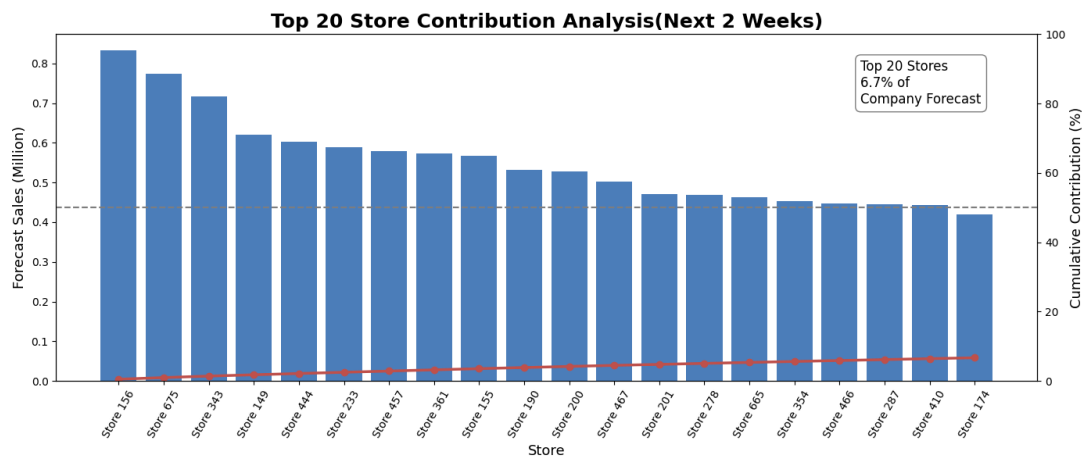


Fig. 13 Top 20 Store Contribution

The next feature presents the weekly sales forecasts for the five highest performing stores. These stores maintain consistently high sales volumes throughout the forecasting horizon while exhibiting a clear weekly sales cycle. Since the COO primarily focuses on the first two forecast weeks, these forecasts provide valuable information for planning replenishment activities, optimising logistics routes, and scheduling sufficient staff during expected periods of high demand.

Furthermore, The 11 missing observations occurred after the required 42 day forecasting horizon and therefore finally did not affect the final 6 week forecast. They were nevertheless imputed to keep the complete future dataset internally consistent.

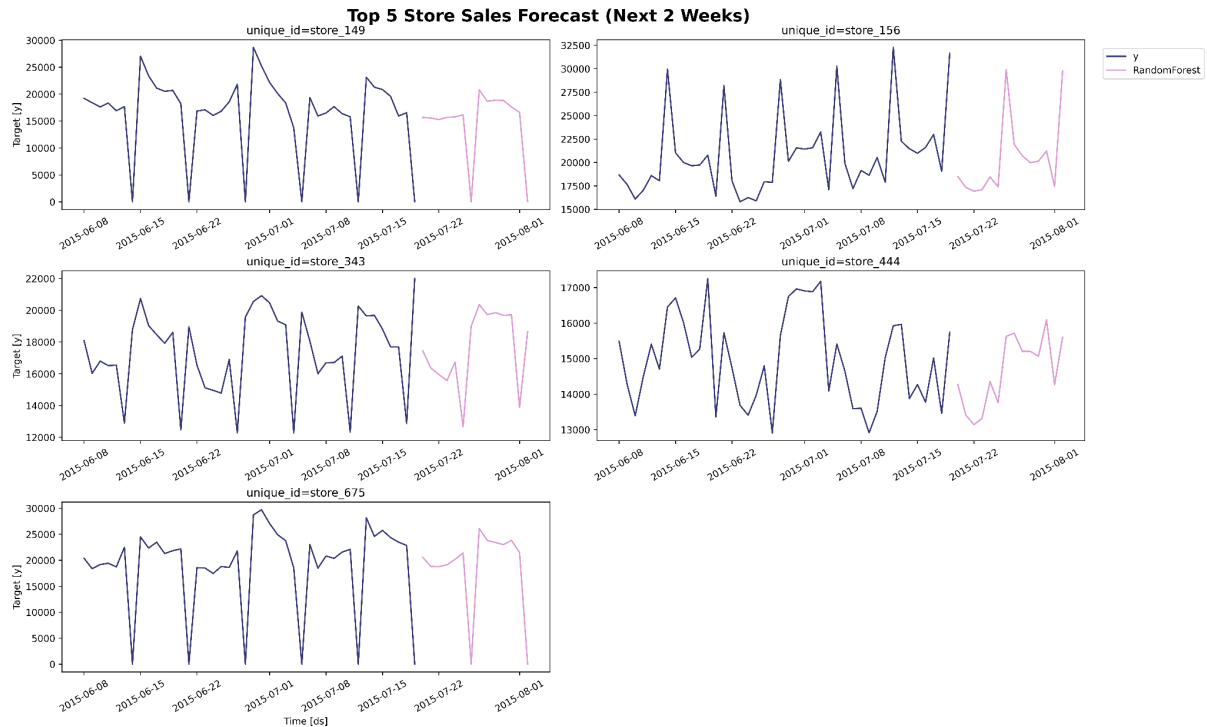


Fig. 14 Top 5 Stores Sales Forecast

To support operational planning at a broader level, forecasts were also aggregated by store type. Clear differences are observed across store formats, with Type A stores consistently generating the highest forecast sales, whereas Type B stores contribute substantially less. Such aggregated forecasts enable managers to allocate operational resources according to store characteristics rather than treating all stores identically. High volume store types may therefore receive priority for inventory allocation and distribution capacity, while smaller store types can adopt more flexible replenishment strategies.

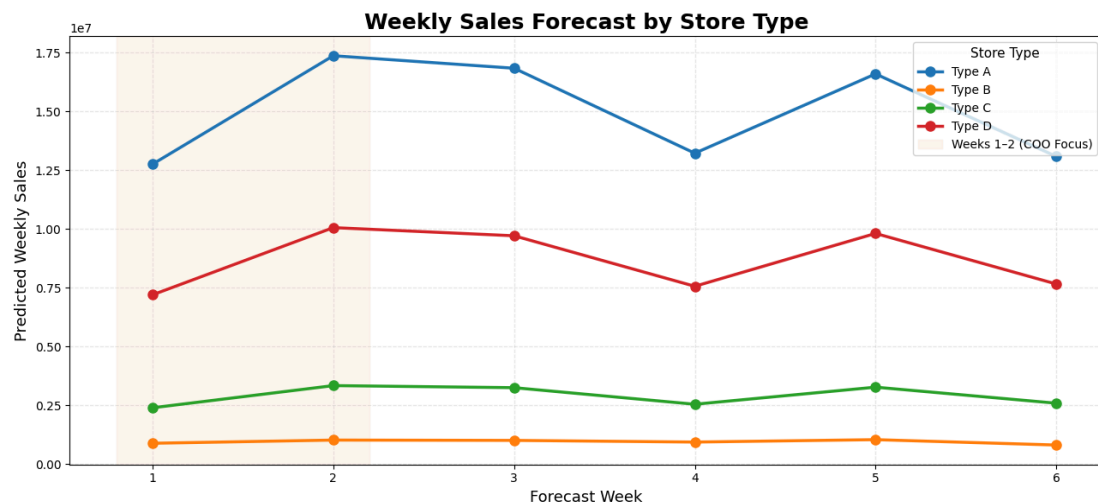


Fig. 15 Store Type Weekly Forecast

Overall, the proposed forecasting framework supports operational decision making at multiple levels. Forecasts for key stores enable targeted resource allocation, while store type forecasts facilitate company wide operational planning. Together, these results provide

practical support for inventory management, logistics coordination, and workforce scheduling during the critical first two forecast weeks.

8. Discussion

8.1 The Reason We Select Global Random Forest

A Global Random Forest was selected as the final forecasting model because it provides a good balance between forecasting accuracy and practical applicability. Compared with the baseline model, incorporating customer forecasts reduced the MAPE from 10.12% to 7.82%.

Unlike local models, the Global Random Forest is trained on data from all stores simultaneously, enabling it to learn shared sales patterns while requiring only a single model to maintain. In addition, it captures nonlinear relationships among historical sales, customer forecasts, promotions, and store characteristics, while feature importance analysis provides useful business interpretability.

Overall, the Global Random Forest offers a robust and practical solution for large scale retail sales forecasting.

8.2 Complexity V.S Maintainability

The proposed forecasting framework balances model complexity and maintainability. Only a small set of Random Forest hyperparameters was tuned using Rolling Forecast Origin Cross Validation, keeping computational costs low while achieving strong forecasting performance.

The feature engineering process captures the main business drivers of sales, including temporal, customer related, promotional, and store features. Therefore, future maintenance is expected to focus mainly on periodic model retraining rather than extensive feature redesign.

Finally, maintaining a single Global Random Forest model is considerably simpler than managing separate models for individual stores, making the framework practical for long term deployment.

9.Maintenance Plan

To maintain forecasting performance, the Global Random Forest model should be retrained weekly as new sales data become available. Since the model is computationally efficient and uses a fixed feature engineering pipeline, weekly retraining requires little maintenance effort while allowing the model to continuously learn the most recent sales patterns.

Forecasting accuracy should also be monitored weekly using MAPE and MAE. If a noticeable deterioration in performance is observed, further error analysis should be conducted before modifying the model.

In contrast, EDA and Permutation Feature Importance analysis are recommended monthly, as they are intended to identify long term changes in sales patterns and feature importance rather than short term fluctuations. If customer related features become less influential or new business drivers emerge, the feature engineering process should be reviewed accordingly.

Finally, data quality, including missing values, outliers, and newly opened stores, should be checked before each weekly retraining to ensure reliable model inputs and maintain long term forecasting performance.

10.Limitations and Future Improvements

Although the proposed forecasting framework achieved satisfactory performance, several limitations remain and provide opportunities for future improvement.

First, customer information is an important predictor of sales, as reflected by the Random Forest models developed with and without customer features. However, future customer numbers are unknown at the forecasting stage and must themselves be predicted. Any error in customer forecasting may propagate to the final sales prediction. If longer historical customer data become available, a more robust customer forecasting model could first be developed and then incorporated into the sales forecasting framework, reducing uncertainty in the final predictions.

Second, the available metadata describing each store is relatively limited. The current model mainly uses store type, competition distance, promotion variables, and competition opening information. For example, competition is represented only by the distance to the nearest competitor. This variable cannot fully describe the competitive environment surrounding each store. Future studies could incorporate richer location information, such as competitor density, urban versus rural characteristics, or local population information, allowing the model to better capture differences in market conditions across stores.

Third, the benchmark analysis shows that forecasting performance differs across stores. Although ETS achieved the best overall performance, it was not the best model for every store. Mean and AutoARIMA still outperformed ETS for a proportion of stores, indicating that stores exhibit heterogeneous time series patterns.

Benchmark models and the final Random Forest models were compared on the same final 42 day holdout period. Rolling forecast origin validation was additionally used for Random Forest hyperparameter selection. While this provides a consistent evaluation framework, future work could further examine model robustness under different forecasting horizons or multiple rolling evaluation periods to assess the stability of forecasting performance under varying business scenarios. Also we could investigate store segmentation or model selection strategies, allowing different forecasting models to be applied to different groups of stores instead of relying on a single model for all stores.

Finally, although Random Forest achieved excellent forecasting accuracy, its predictions are less interpretable than traditional statistical models. While permutation feature importance provides useful insights into the contribution of each feature, it cannot fully explain individual predictions. Future work could explore more advanced explainability methods to improve model transparency and support business decision making.

Moreover, due to the request of COO, who is interested in accurate weekly forecasts for the next 2 weeks, also accuracy for 6 weeks can conduct an overall plan for the company, we should do more tests for accuracy in 2 weeks or even develop a professional model for this forecast.

AI Declaration

Used ChatGPT and Codex for code fixation and recommendation, better coding grammar way to approach the same goal and tidying up csv. production. We finally identified the result by ourselves.

Used ChatGPT for fixation and checking the indentation, consulting whether our solution is practical in real world for some steps and checking English grammar.

Note for Code Execution

Since **AutoARIMA** takes a very long time to run for all stores, we prepared two versions of our notebook.py on moodle for convenience:

Yu Ching Lin: The benchmark models are placed in the middle of the workflow. This version presents the complete modeling process in chronological order and is intended to clearly demonstrate our full methodology.

Xueying Peng: The benchmark section is moved to the end of the workflow. This allows the Random Forest model to be executed directly without waiting for AutoARIMA to finish, making it much practical for reviewing the machine learning part.

Both notebooks produce the same final results and contain the same analyses; the only difference is the order of execution to reduce runtime during evaluation.